DeepCurvMRI: Curvelet Transform-Based Deep Learning for Early Detection of Alzheimer's Disease

Paper Overview

Topic: Early detection and multi-class staging of Alzheimer's Disease from structural MRI using a Curvelet Transform-based feature extraction pipeline combined with a lightweight, from-scratch Convolutional Neural Network.

This summary examines a 2023 IEEE Access study that proposes DeepCurvMRI, a diagnostic pipeline that converts raw MRI slices into Fast Discrete Curvelet Transform (FDCT) coefficients, denoises them using a kurtosis-based thresholding scheme, and classifies the resulting representation with a compact CNN. The model targets both binary (Non-Demented vs. Very Mild Demented) and four-class (Non-Demented, Very Mild, Mild, Moderate Demented) staging of Alzheimer's disease.

Paper Title	Author(s) / Year	Dataset / Source	Model / Technique	Key Findings
DeepCurvMRI: Deep Convolutional Curvelet Transform-Based MRI Approach for Early Detection of Alzheimer's Disease	Chabib, Hadjileontiadis & Al Shehhi (2023)	Kaggle Alzheimer's MRI Dataset (4-class): 6,400 images from 200 subjects (32 slices each) — ND, VMD, MID, MOD classes.	Fast Discrete Curvelet Transform (FDCT, wrapping method) for feature extraction + kurtosis-based denoising thresholding + a shallow custom CNN (2 conv layers, batch norm, max-pooling, GAP, dropout, dense, softmax).	Achieved **98.62% multi-class accuracy** (F1 = 99.21%) via leave-one-group-out CV, and **98.71% binary (ND vs. VMD) accuracy**. Outperformed VGG-16 (+28.4%) and AlexNet (+4.6%) while using only ~51,797 parameters — far fewer than comparison models.

Summary

The analyzed study converges on three core technical layers needed to build an effective MRI-based multi-class diagnostic system for Alzheimer's Disease:

1. **Curvelet-Based Feature Extraction:** Standard MRI scans are decomposed using the Fast Discrete Curvelet Transform, which — unlike wavelet transforms — represents both linear and curved edges compactly across multiple scales and orientations. Only the coarse scale (17×17) is retained, since higher scales mainly add noise given the dataset's low native resolution.
2. **Kurtosis-Based Denoising:** Curvelet coefficients are thresholded using a kurtosis-weighted scale-dependent threshold (extending Donoho & Johnstone's method), which strips coefficients dominated by non-Gaussian noise while preserving anatomically meaningful edges — yielding a 0.35% accuracy gain over unweighted thresholding.
3. **Lightweight CNN Classification:** Rather than a heavy pretrained backbone, the study trains a shallow CNN from scratch (two conv layers, batch normalization, max-pooling, global average pooling, dropout, dense + softmax) directly on the thresholded curvelet representation, reaching a peak multi-class accuracy of 98.62% and F1-score of 99.21% with roughly 51,797 parameters — orders of magnitude fewer than VGG-16 or AlexNet.

Limitations & Gaps:

The primary limitation is severe class imbalance in the dataset (only 64 Moderate Demented images vs. 3,200 Non-Demented), which the paper does not explicitly address beyond cross-validation. Data leakage is a further concern:

since each subject contributes 32 slices, subject-wise leave-one-group-out validation is essential but was only rigorously applied here (unlike some comparison methods). The pipeline is also evaluated on a single Kaggle dataset without external/clinical validation, and the curvelet + kurtosis-thresholding preprocessing is a multi-stage, hand-engineered process that is less end-to-end than pure deep-learning alternatives, and it lacks explicit attention or explainability mechanisms to localize the atrophy regions driving each prediction.